DATABASE LAB @POSTECH

From LILaC and FiF to Real-World Multimodal Retrieval

2026.2.12

Presenter: **Joohyung Yun**

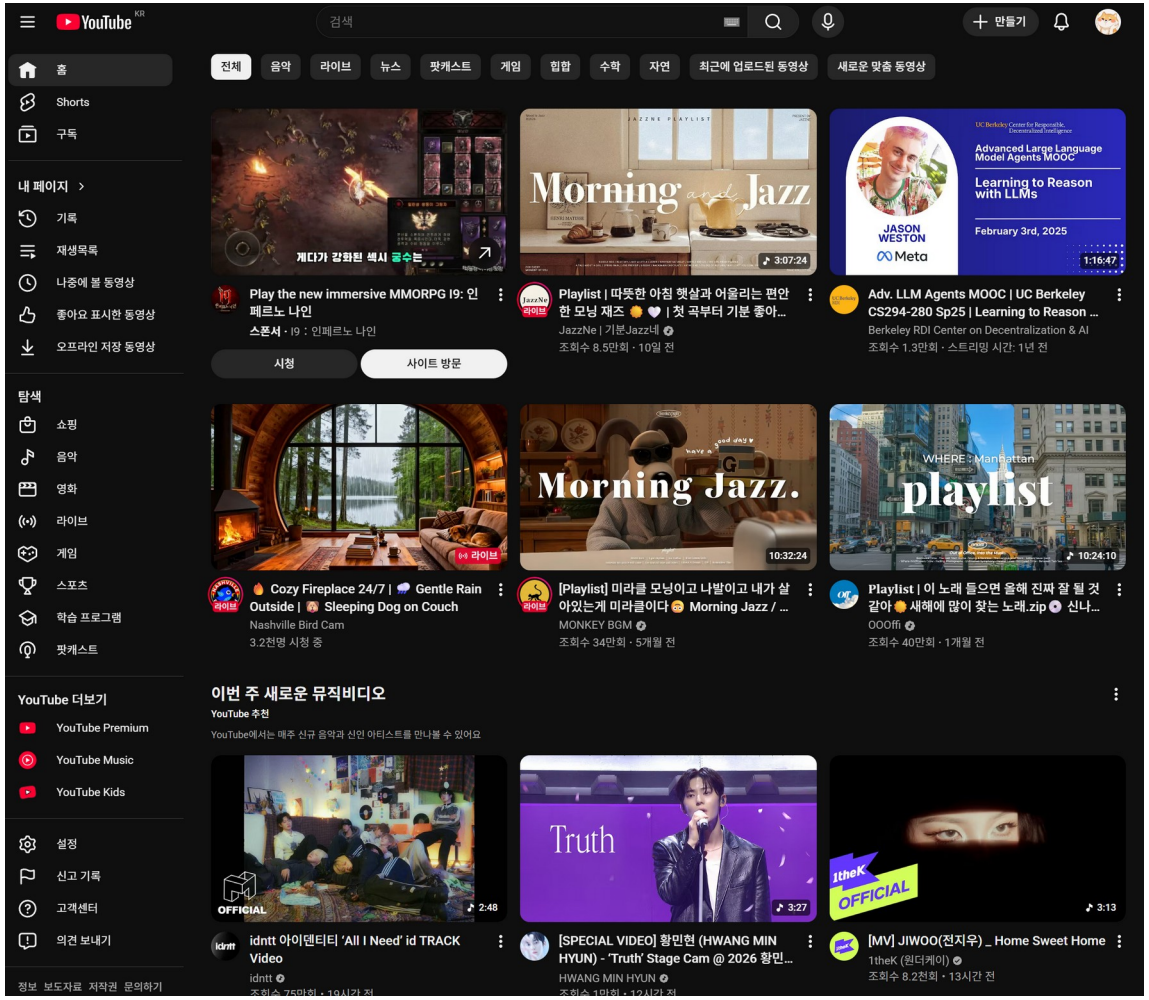
Overview of This Seminar

- Top Goal of My Research
- Brief Introduction of My Previous Work
- How Does Real-World Multimodal Retrieval Differ
- Future Challenges to Solve
- Viable Research Topic Options

Retrieving Info From Webpages Matter

- The web is the **largest and most up-to-date knowledge source** — but also the noisiest and most dynamic.
- Next-gen AI products (search, assistants, copilots) require retrieval that works on **real webpages, not curated benchmarks**.
- **Multimodal evidence** is everywhere on the web: text + tables + figures + UI elements + (increasingly) video/audio.

My long-term goal: make retrieval robust enough for real-world, open-web data.



AI Entrepreneurs Want “Web-Native” Retrieval

Goal **Solve information retrieval in the open web at production-grade constraints:**

- Input is raw HTML + rendered pages
- Pages are dynamic/reactive (scroll, expand, login, infinite feed)
- Multi-modality expands (audio/video/UI interactions)
- Tight latency/cost budgets + reliability (provenance, robustness)

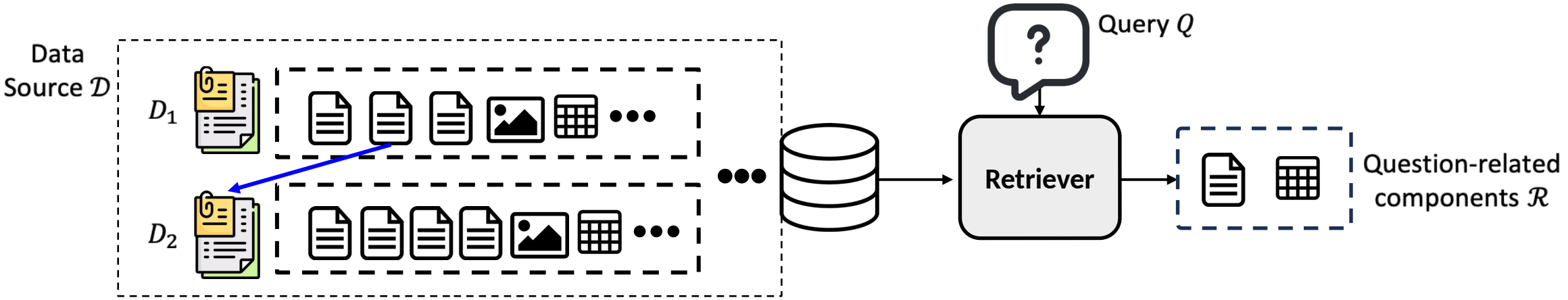
Brief Introduction of My Previous Work

A Relaxed Problem Setting

Open-Domain Multimodal Document Retrieval

Given a data source $\mathcal{D}$, a link mapping $\mathcal{L}$ and a query Q, correctly retrieve a ranked list of components $\mathcal{R} = [C_1, \ldots, C_{n_{ret}}]$. $\mathcal{R}$ should contain the ground truth set of components $\{C_{gt_1}, \ldots, C_{gt_r}\}$.

- Data source $\mathcal{D} = \{D_1, \ldots, D_{k_{doc}}\}$
- Document $D = \{C_1, \ldots, C_{k_{comp}}\}$
- Component $C \in \{P, T, I\}$
  - P: paragraph, T: table, I: image
- Link mapping $\mathcal{L} = \mathcal{C} \rightarrow \mathcal{D}$
  - $\mathcal{C}$: A set of all components within $\mathcal{D}$



LILaC (2025): Embedding-based Traversal on Graph Structure

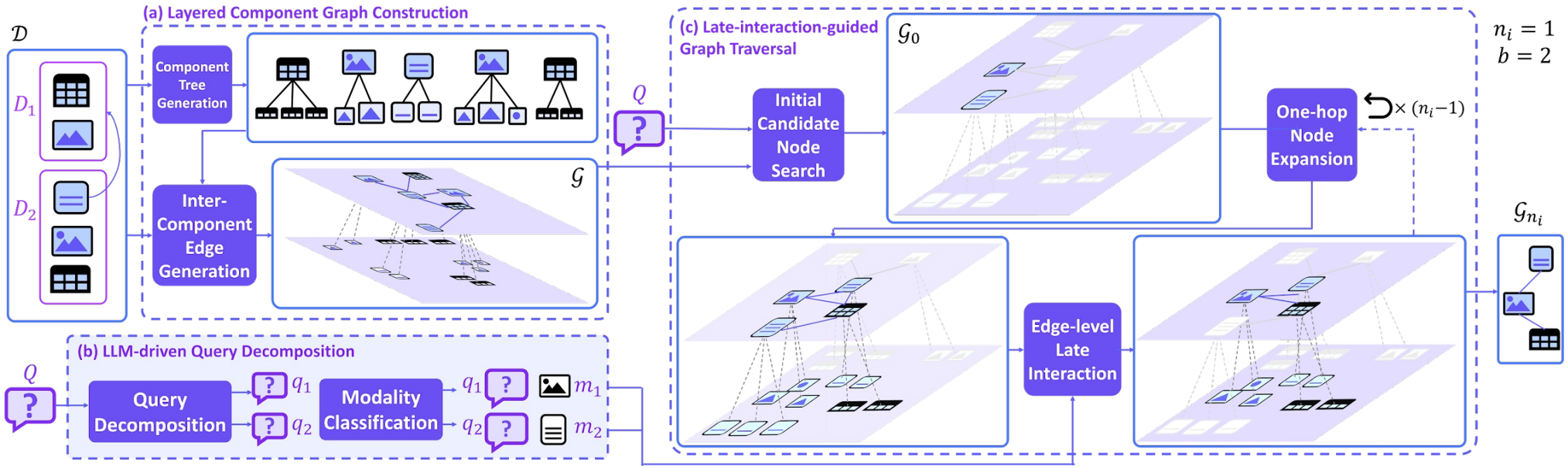
Core idea **Organize documents into a 2-layer graph and traverse edges with late interaction scores.**

Enables

- Coarse-to-fine evidence
- Multihop via navigational links
- Edge scoring with late interaction

Key assumptions

- Offline graph + embeddings
- Pre-extracted components/subcomponents
- Reliable hyperlink mapping



FiF (2026): Failure-aware, Cost-aware Agentic Traversal

Core idea

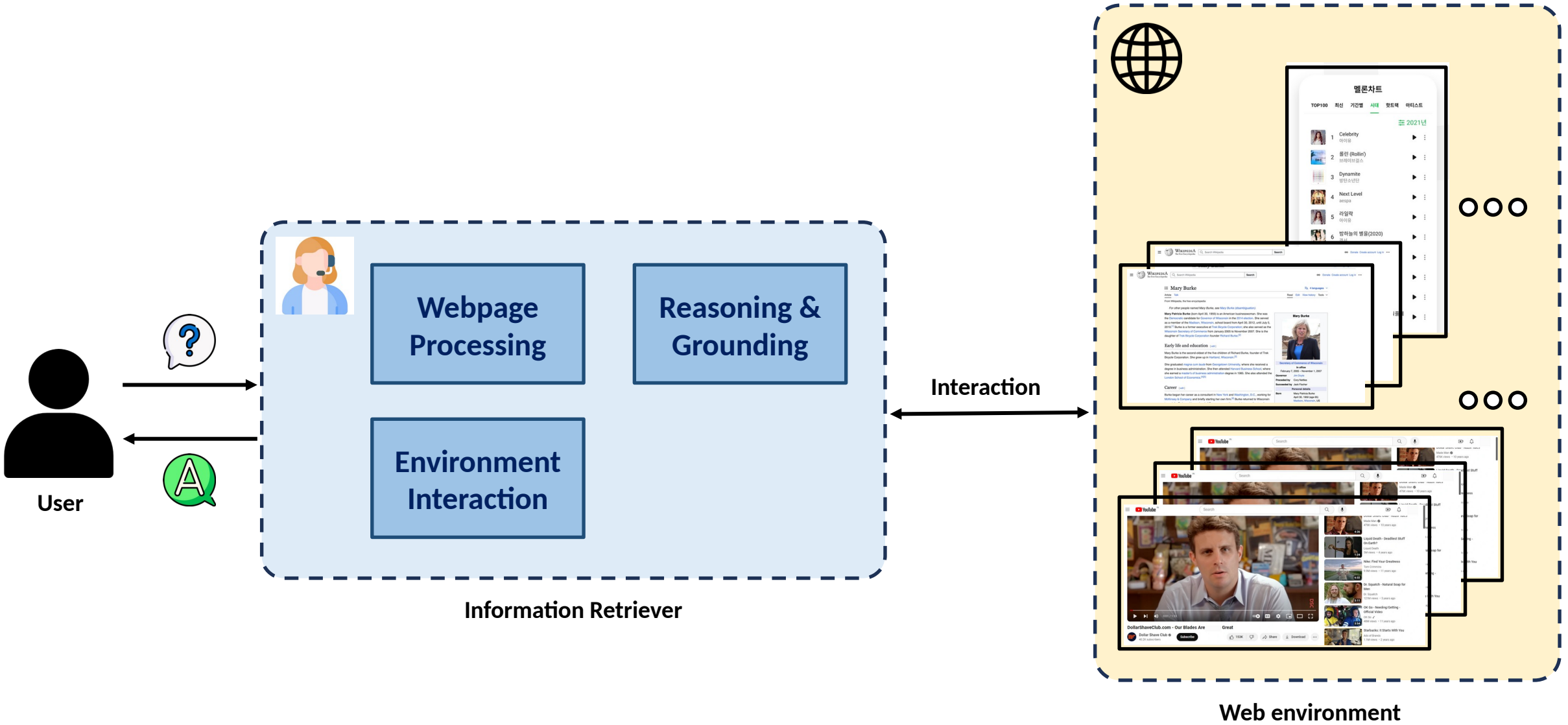
Treat multihop retrieval as an information-state MDP: revise subqueries, switch strategies, and backtrack using failure traces.

Still assumes

- Clean component graphs + reliable navigational signals
- Static pages (no reactive UI / hidden content)
- Offline corpora with URL annotations

What Should be Considered Additionally for a Practical Real-World Multimodal Retrieval?

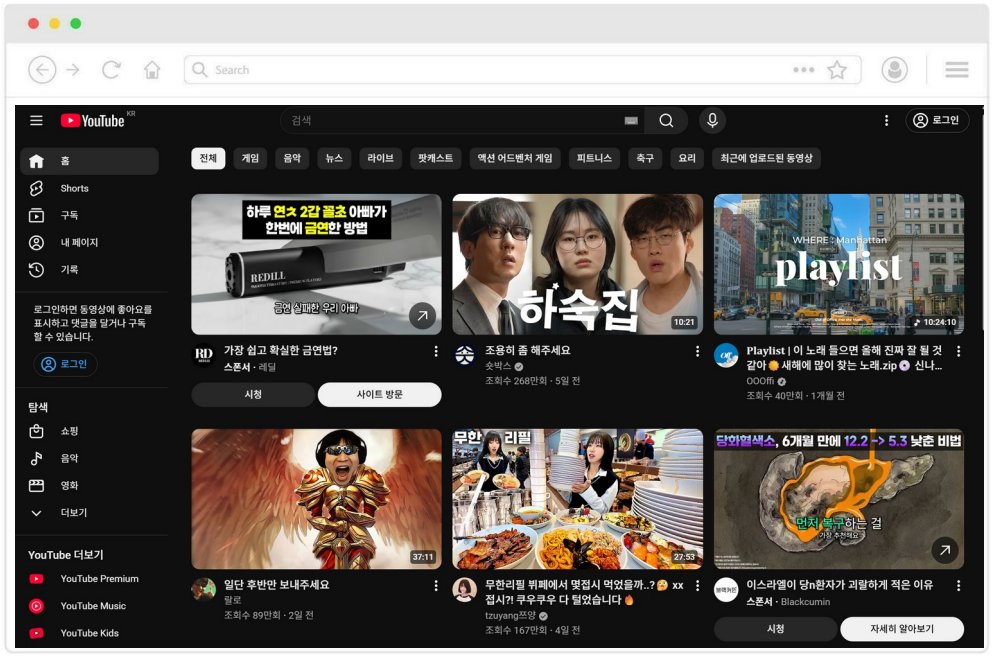
Diagram of Practical Real-World Information Retriever



Real Web Input ≠ Preprocessed Components

The retriever should start from raw web

- Input is raw HTML/DOM + CSS/JS + rendered pixels (not clean paragraphs/tables/images).
- Need end-to-end extraction: boilerplate removal, component segmentation, table parsing, figure/plot understanding.



Raw web



Immediate outputs



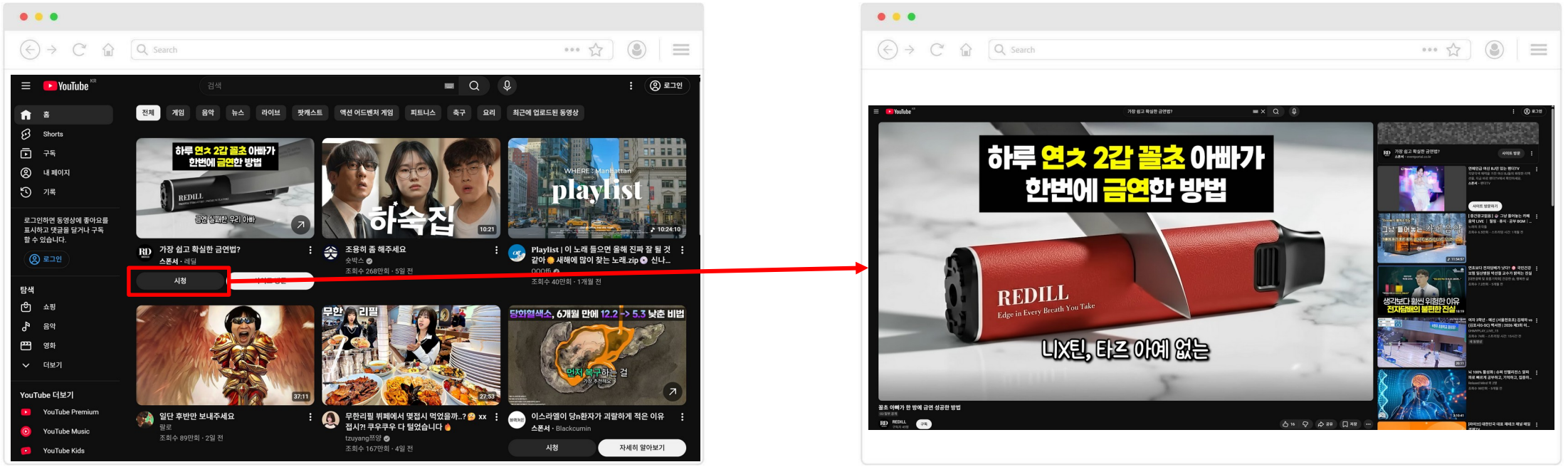
Processed components

More Modalities: Beyond Text, Table and Image

Multi-modality

- Video: retrieve relevant time spans and frames
- Audio: retrieve segments and transcripts.

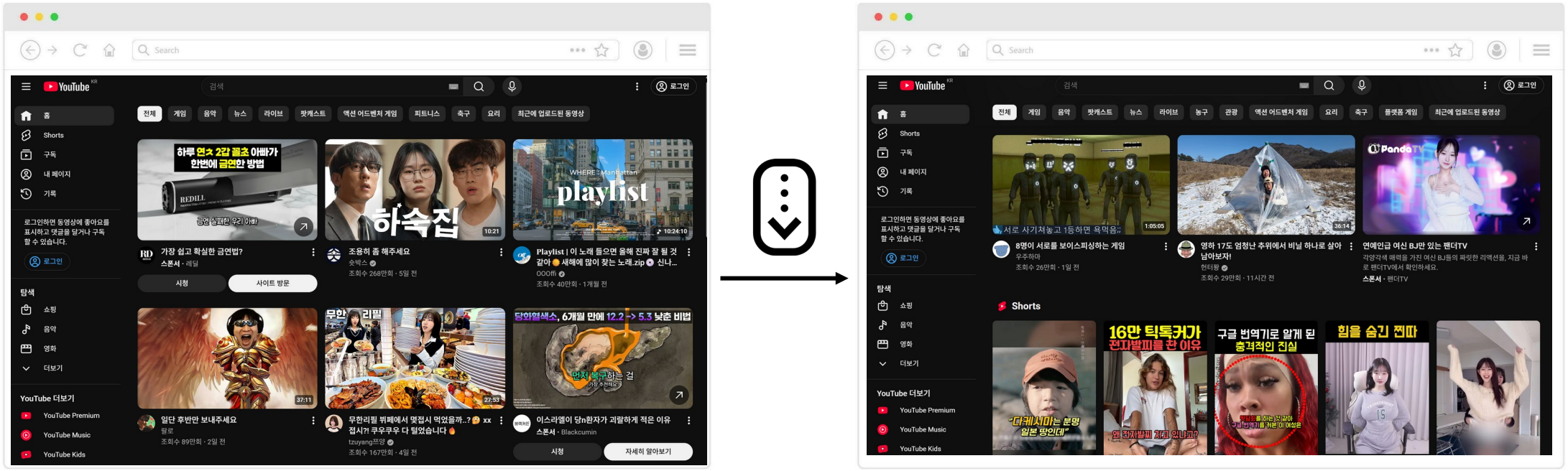
UI signals: buttons, menus, tooltips, and layout semantics can carry meaning.



Dynamic and Reactive Webpages

Evidence may be hidden behind actions

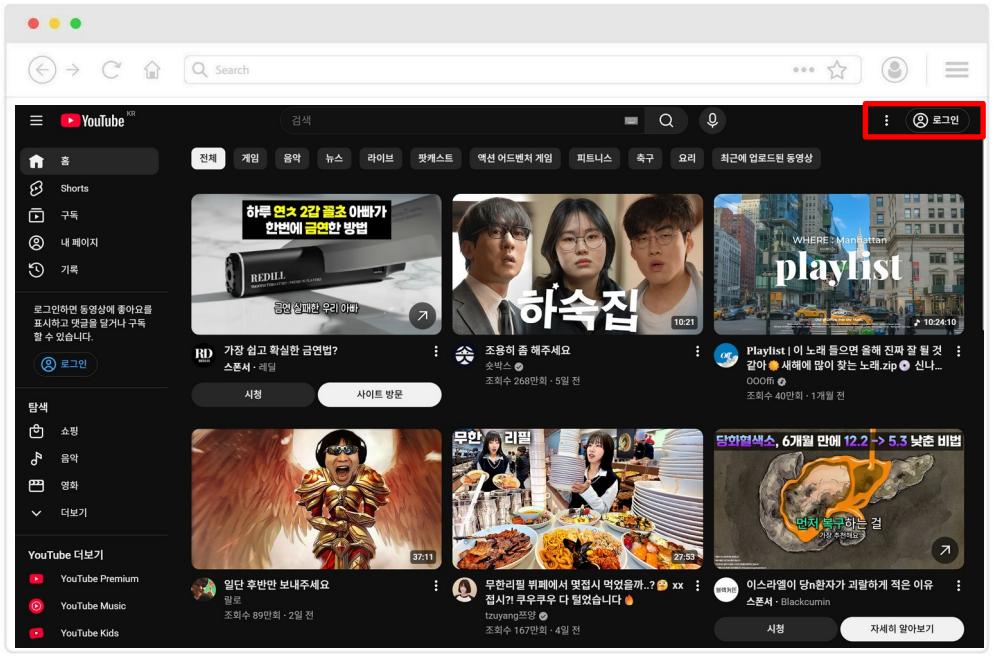
- Content appears after scroll, expand, tab switch, pagination, login, or async requests.
- Retrieval becomes stateful: observe → act → re-observe → retrieve



Challenges in Offline Indexing

The corpus changes continuously and is just too big

- Indexing should be done incremental/streaming, for the entire web environment.
- Pages update, disappear, and personalize.





Web environment

Grounded Component Retrieval

Retrieve evidence with grounding

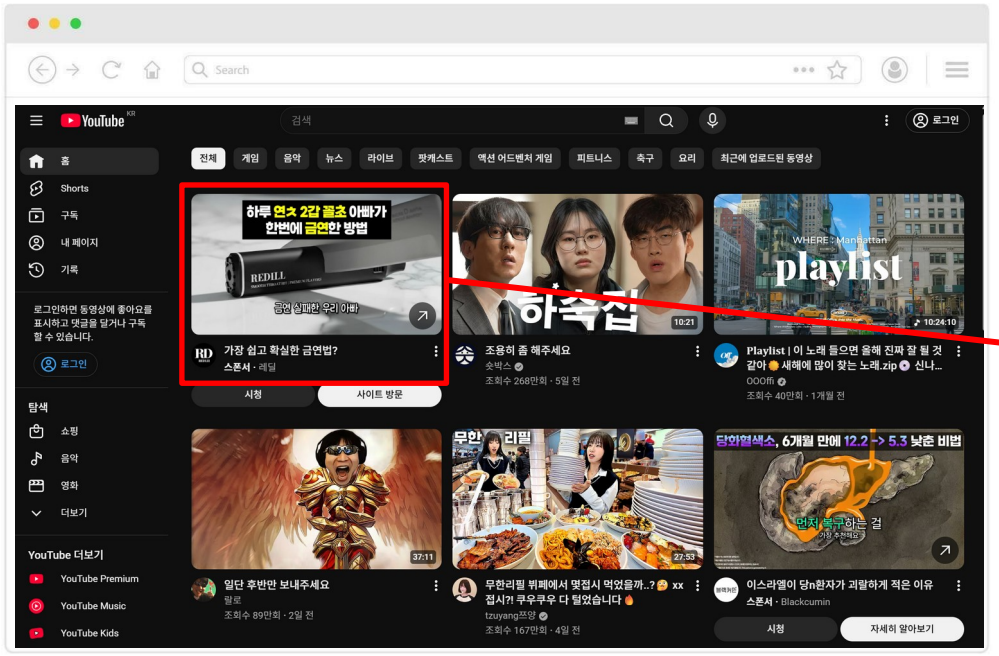
- Handle multimodal grounding: cite the exact table rows / chart region / image crop used.

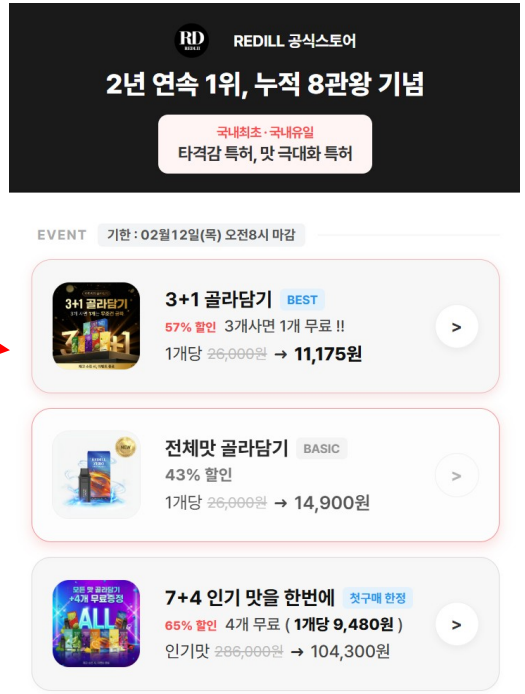


Production Constraints

Enterprise-grade budgets and reliability

- Latency/cost budgets: avoid full-page LLM/OCR per request; use adaptive pipelines and caching.
- Robustness: withstand spam/SEO, templates, duplicates, and noisy link signals.
- Compliance: robots.txt, access control, PII, copyright; must support safe failure modes.



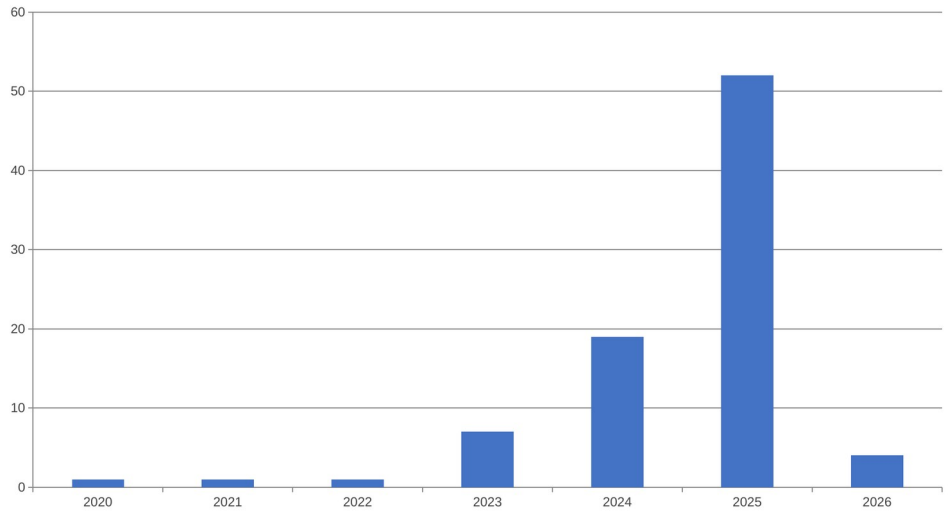


Current Work & Their Challenges in Solving Real-World Multimodal Retrieval

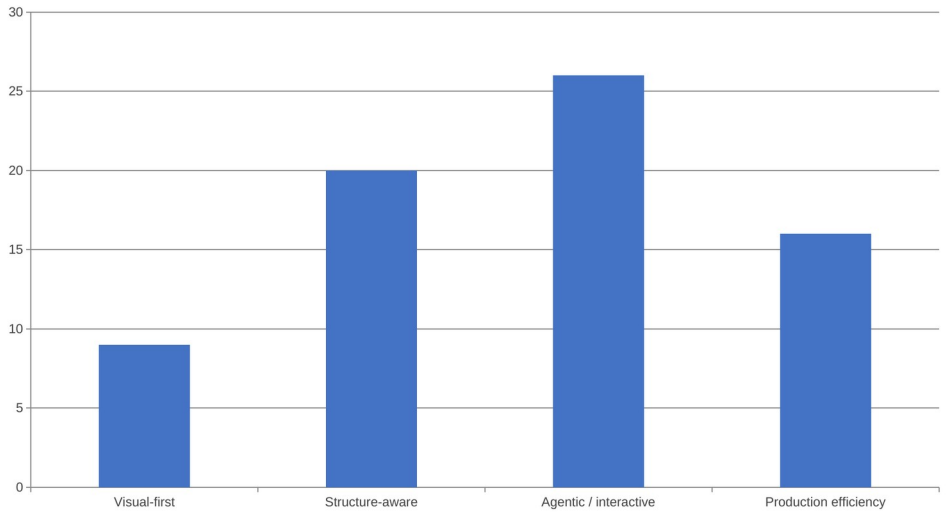
How the Research Community Tackles “Web Realism”

Survey: 85 papers (2020–2026). Tags are non-exclusive (a paper can belong to multiple directions).

Papers Published per Year



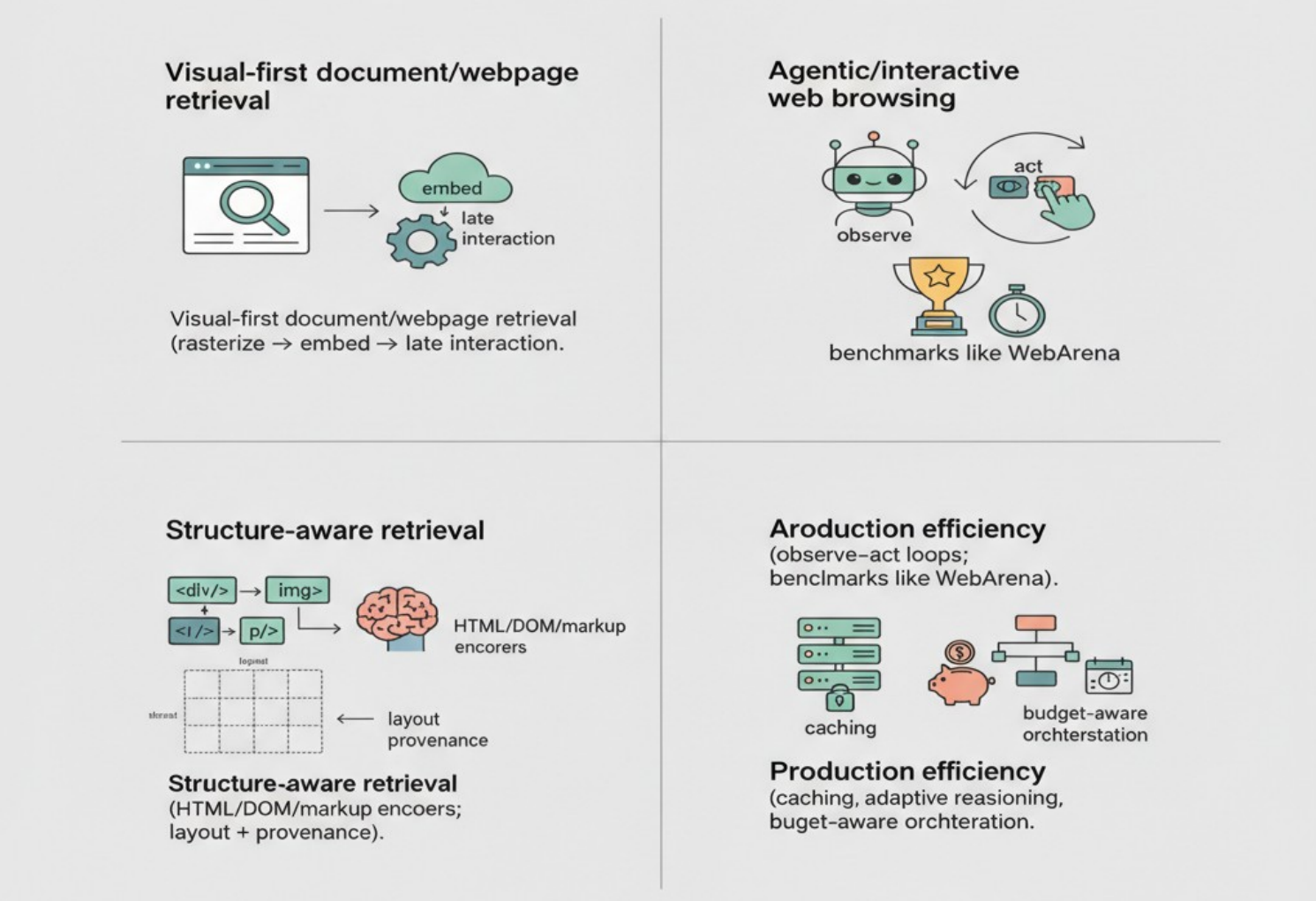
Papers Published for Each Direction



Venue	# papers	Share
arXiv	43	50.6%
NeurIPS / ICLR / ICML	18	21.2%
ACL / EMNLP / NAACL	16	18.8%
ACM MM	5	5.9%
SIGIR / TPAMI	2	2.4%
Other	1	1.2%

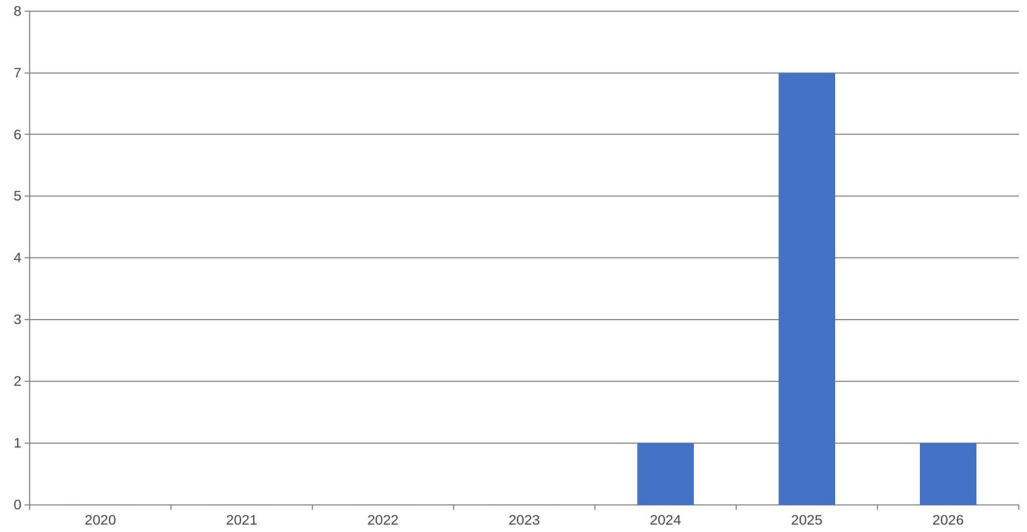
How the Research Community Tackles “Web Realism”

Four directions emerging across ML/NLP + industry



Visual-Embedding Retrieval

Papers tagged
9



Venue	#	Share
arXiv	6	67%
NeurIPS / ICLR / ICML	1	11%
ACL / EMNLP / NAACL	1	11%
ACM MM	1	11%

Typical approach

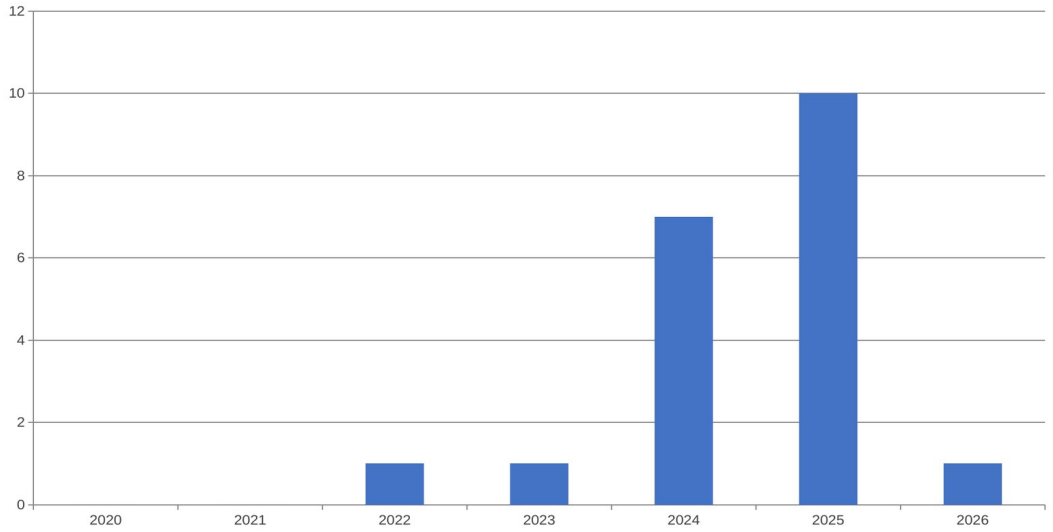
- Rasterize webpages (full page / regions) into VLM embeddings.
- Exploit layout as a signal, often reducing dependence on brittle HTML parsing.
- Use OCR/region proposals when grounding is required.

Open gaps

- Region alignment: align pixel regions back to DOM nodes and page states.
- Cost: rendering/OCR is expensive; need caching + on-demand execution.
- Robustness to templates, ads, and responsive layouts.

Structure-Aware Retrieval

Papers tagged
20



Venue	#	Share
arXiv	10	50%
NeurIPS / ICLR / ICML	6	30%
ACL / EMNLP / NAACL	2	10%
ACM MM	1	5%
SIGIR / TPAMI	1	5%

Typical approach

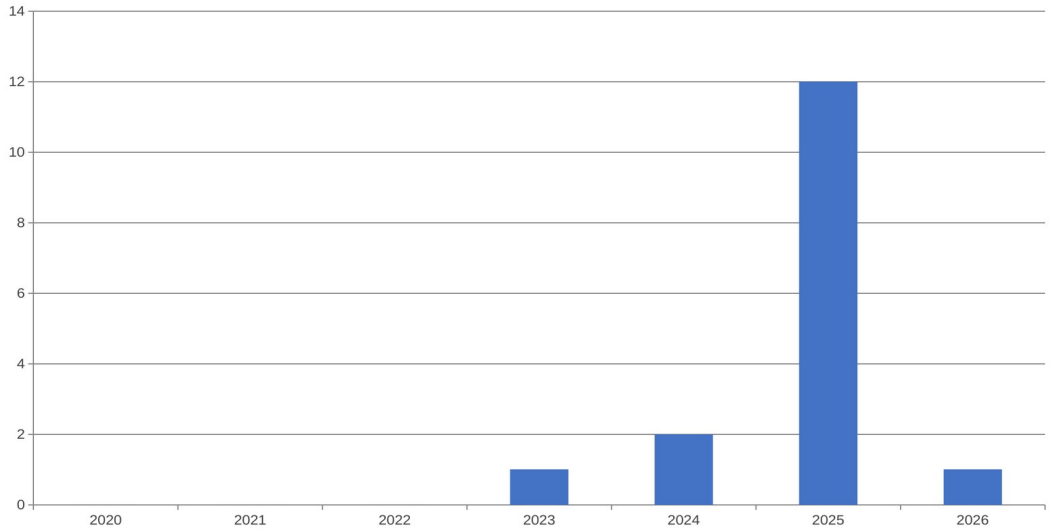
- Encode HTML/DOM/markup with structure-aware Transformers; index blocks/nodes as retrieval units.
- Use DOM hierarchy, attributes, and link structure to complement text matching.
- Fuse structure with visual/layout cues for better grounding.

Open gaps

- End-to-end segmentation under noisy templates & JS-generated DOM.
- Joint modeling of DOM + pixels + cross-page navigation signals.
- Standardized evaluation for grounded evidence and dynamic behavior.

Production Efficiency

Papers tagged
16



Venue bucket	#	Share
arXiv	9	56%
NeurIPS / ICLR / ICML	3	19%
ACL / EMNLP / NAACL	2	12%
SIGIR / TPAMI	1	6%
Others	1	6%

Typical approach

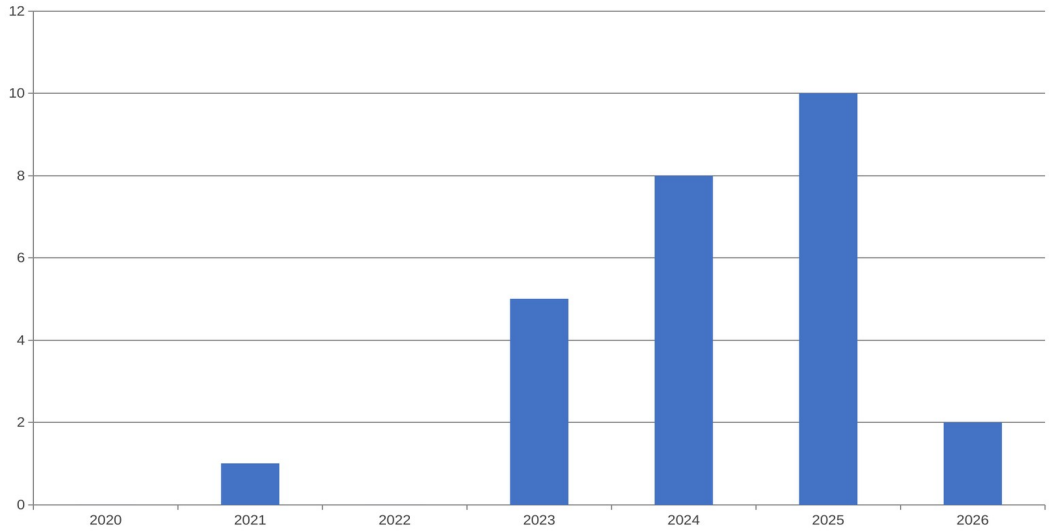
- Caching (embeddings, KV-cache), adaptive compute, and selective recomputation.
- Streaming updates + freshness-aware ranking for changing corpora.
- System-level evaluation that includes latency and cost.

Open gaps

- Unified optimization across retrieval + generation + browsing actions under SLOs.
- Benchmarks for realistic workloads (long tail, partial failures, distribution shifts).
- Compliance/safety constraints integrated into end-to-end pipelines.

Agentic Web Browsing

Papers tagged
26



Venue	#	Share
arXiv	11	42%
NeurIPS / ICLR / ICML	7	27%
ACL / EMNLP / NAACL	6	23%
ACM MM	2	8%

Typical approach

- Formulate retrieval as sequential decision-making in browser environments (observe–act–retrieve).
- Observation pruning / line retrieval to control context and costs.
- Backtracking and tool use to recover from dead-ends.

Open gaps

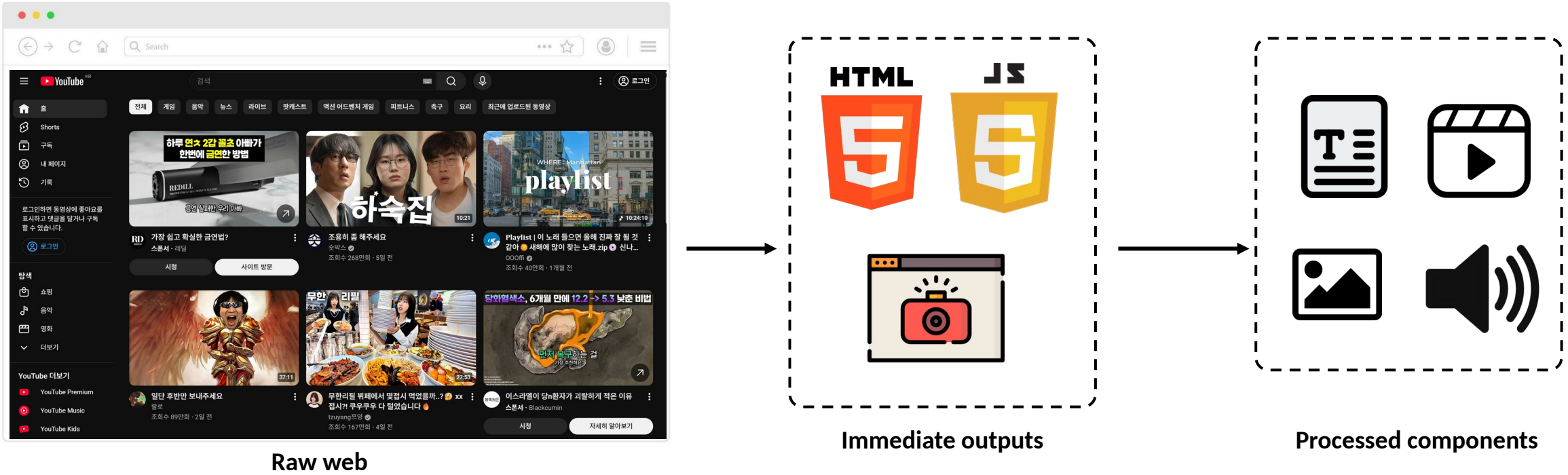
- Multimodal evidence retrieval in the loop (tables/figures/video) with grounding.
- Budgeted control (latency/cost) with reliability guarantees.
- Generalization across sites + adversarial/SEO noise.

What Should We Research Next?

Topic 1: Web-Native Processing Within Retrieval

End-to-end: HTML/DOM + render → components → grounded retrieval

- Goal: remove the “pre-parsed components” assumption in LILaC/FiF.
- Key research: robust componentization under noise + provenance-aware ranking.



Topic 2: Reactive Web Retrieval

Retrieval + actions on dynamic pages (observe–act–retrieve)

- Extend FiF's tool list to atomic real web actions (scroll/click/login).
- Learn when to explore vs. exploit, and when to backtrack or re-anchor.

Topic 3: Modality Extension

Extend components to video/audio/temporal spans

- Retrieve: video time ranges, keyframes, audio segments, UI-state evidence.
- Need temporal grounding + scalable indexing for long media.

Topic 4: Freshness-Aware Web IR under Budgets

Streaming indexing + caching + adaptive computation

- Move from offline embedding to hybrid (precompute + on-demand) pipelines.
- Optimize accuracy–freshness–cost jointly (enterprise SLOs).
- Deliverables: retriever with incremental updates and cache-aware routing.